

Homework 1

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1. **Claim:** $\sum_{v \in G} \deg(v) = 2|E|$ for undirected graphs, and $|E|$ for directed graphs.

Proof. In an undirected graph, if (u, v) is an edge, both $\deg(u)$ and $\deg(v)$ count that edge. This means that when we compute the sum $\sum_{v \in G} \deg(v)$, we actually double count each edge. Thus, the sum is equal to $2|E|$.

In the case of directed graphs, we no longer double count each edge because $(u, v) \neq (v, u)$. Thus, in directed graphs, we only count each edge once, so the sum equals $|E|$. $\square$

2. (a) **Claim:** An undirected graph has an even number of vertices of odd degree.

Proof. Split the vertices into two groups: vertices that have an odd degree (set X) and vertices that have an even degree (set Y). We know that $\sum_{v \in Y} \deg(v)$ is always even, and $\sum_{v \in X} \deg(v)$ is even iff $|X|$ is even. Since the Handshaking Lemma requires the total sum of all degrees to be $2|E|$, which is even, we need $|X|$ to be an even number. $\square$

- (b) **Claim:** If a vertex v has odd degree in the graph G then there exists another vertex w , also of odd degree, such that v and w are connected by a path.

Proof. Let v be a vertex with an odd degree. Furthermore, let C be a connected component of G that contains v . Assume on the contrary that v is the only vertex with odd degree in C . By the Handshaking Lemma, we must have $\sum_{u \in C} \deg(u)$ be even. However, since v has an odd degree and all the other vertices have even degree, this sum is odd, a contradiction. Therefore, there must be another vertex $u \in C$ such that u has odd degree. Furthermore, since $v, u \in C$, there is a path from v to u . $\square$

3. (a) **Graph of number pairs.**

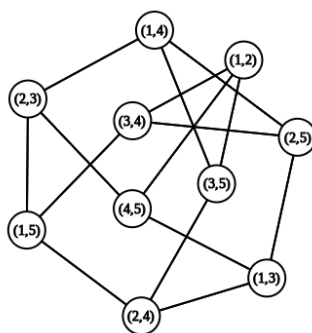


Figure 1: Graph of number pairs

(b) **Claim: The graph has 15 edges.**

Proof. Each vertex can be connected to 3 other vertices. Since there are $\binom{5}{2} = 10$ vertices, this creates 30 possible edges. However, this double counts each edge since it is an undirected graph, so we divide by 2 to get our final answer of 15 edges. $\square$

4. (a) **Claim: There are 2^n vertices.**

Proof. Note that each string of bit string corresponds to one unique vertex. There are 2 choices for each bit, and there are n bits in a string, so there are 2^n strings, and thus 2^n vertices. $\square$

(b) **Claim: Each vertex has n neighbors and each edge has $2n - 2$ neighbors.**

Proof. Given a vertex's bit string, you can get all its neighbors by changing exactly one bit in the string. Since there are n places to change the bit, each vertex has n neighbors.

Each edge connects two vertices u and v . Each vertex has $n - 1$ other edges connected to it (since there are n neighbors for each vertex). This means that each edge has $2(n - 1) = 2n - 2$ neighbors. $\square$

Proof. Since there are 2^n vertices and each vertex is connected to n other vertices through n edges, there are $n2^n$ total vertices. However, we've double counted each edge since the direction doesn't matter, so we divide by 2 to get $n2^{n-1}$. $\square$

Proof. Since each vertex is on a line, the furthest you would travel would go from vertex 1 to vertex n . This would go through $n - 1$ edges, so the diameter is $n - 1$. $\square$

Proof. The furthest you can travel would be to go from one vertex to the vertex opposite it on the circle. This requires you to traverse $\lfloor n/2 \rfloor$ edges, so it is the diameter. $\square$

Proof. Every vertex is connected to every other vertex by an edge, so you only need to travel across 1 edge to reach any other vertex. This makes the diameter 1. $\square$

Proof. The furthest you can travel is from a vertex on the left side to the vertex that lies on the edge that connects the two sides, to the other vertex on that edge, to any edge on the right side. This requires you to travel along 3 edges so the diameter is 3. $\square$

Proof. See Figure 2 of the moves requires. Here, each vertex label represents who is in the boat. $\square$



Figure 2: Graph of valid sequence

7. Graph of the knights problem

1	2	3
4	5	6
7	8	9
10	11	12

Table 1: Chess Board

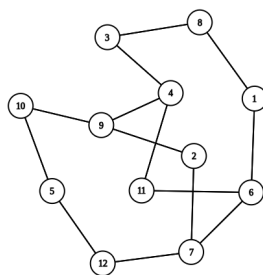


Figure 3: Graph of knight moves

8. Cayley's Group

Note: this is an image I found on the internet [here](#), since I can't render upside-down characters in LaTeX. It's the same graph but it uses the letter F instead of R . Altogether, there are 4 possible rotations and 2 choices for the flip (either its flipped along the x -axis or not), for a total of 8 states.

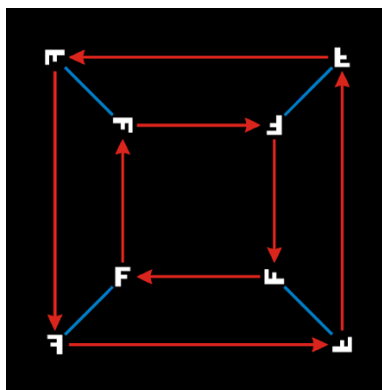


Figure 4: Graph of number pairs

9. (a) **Adjacency matrix for P_5**

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- (b) **Adjacency matrix for C_5**

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

- (c) **Adjacency matrix for K_5**

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

10. (a) **Claim: The sequence is graphic.**

Proof. See Figure 5 for the graph. It is clearly simple, so the sequence is graphic. $\square$

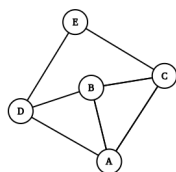


Figure 5: Simple graph from part (a)

(b) **Claim: The sequence is not graphic.**

Proof. Note that the sum of the degrees is $5+4+3+2+1 = 15$, which is odd. This violates the Handshaking Lemma, so the sequence is not graphic. $\square$

(c) **Claim: The sequence is not graphic.**

Proof. Note that there are 5 vertices and two vertices with degree 4. These two vertices connect to every other vertex, which means that every vertex in the graph has a degree of at least 2. This is not possible however because the sequence has a 1, a contradiction. Therefore, the sequence isn't graphic. $\square$

(d) **Claim: The sequence is not graphic.**

Proof. The sum of the degrees is 17, which violates the Handshaking Lemma since it should be even and not odd. $\square$

(e) **Claim: The sequence is graphic.**

Proof. See Figure 6 for the graph. It is clearly simple, so the sequence is graphic. $\square$

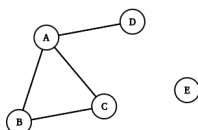


Figure 6: Simple graph from part (e)

11. **Claim: If every vertex of a finite directed graph G has out-degree at least 1, then the graph contains a directed cycle.**

Proof. Choose any vertex $v_1 \in G$. Since it has an out-degree at least 1, you can travel to another vertex v_2 . Since every vertex has an out degree at least 1, you can always travel to another vertex. This forms a chain $v_1, v_2, v_3, \dots$. Finally, since the graph is finite, you must repeat a vertex (that is there will eventually be a vertex v_n that connects to a previously seen v_m) at least once. Thus, you've formed a directed cycle $v_m, v_{m+1}, \dots, v_{n-1}, v_n, v_m$. $\square$

12. **Claim: The diameter of a connected graph is at most $|V| - 1$.**

Proof. It's obvious that you can only travel to $|V| - 1$ other vertices from any starting vertex (traveling to yourself would only require 1 step, which clearly doesn't have to be the maximum for all graphs). Thus, the max steps you can take would be to travel across every other vertex exactly once before hitting the last one, and this would mean traveling across $|V| - 1$ edges. Thus, this is the largest diameter of a connected graph. $\square$

13. **Claim: The betweenness centrality of vertex i is $(i - 1)(n - i)$.**

Proof. Since this is a path graph, there's only one path between any two vertices. Thus, $\sigma_{st} = 1$. We just need to count the number of vertices before vertex i and the number of vertices after vertex i , and then choose one from each group. There are $i - 1$ vertices before vertex i and $n - i$ vertices to the right of vertex i , so there are $(i - 1)(n - i)$ paths. This is the betweenness centrality of vertex i . $\square$

14. (a) **Claim: If there is a walk from u to v then there is a path from u to v .**

Proof. This is true. Let's say the walk from u to v contained some cycles. You'd eventually have to stop traveling along the cycle and move to a new vertex, so in the path, you could just cut out all the cycles, thus forming a path. $\square$

(b) **Claim: If there is a path from u to v and a path from v to u , then there is a cycle that includes u and v IF it's a directed graph. Otherwise, it's not true.**

Proof. For directed graphs, the statement is already true. If $u, a_1, a_2, \dots, v$ is a path from u to v and $v, b_1, b_2, \dots, u$ is a path from v to u , then $u, a_1, \dots, v, b_1, \dots, u$ is a cycle.

However, if it is an undirected graph, then the statement doesn't have to be true. Consider a graph with two nodes 1 and 2 with a single edge connecting them. There is a path from 1 to 2 and from 2 to 1, yet there are no cycles in the graph. $\square$

15. Claim: An undirected graph is bipartite iff it has no cycles of odd length.

Proof. Front: Let X, Y be subsets of the vertices of graph G such that $X \cap Y = \emptyset$ and $X \cup Y = G$. These are the two disjoint set of vertices of graph G . Now, assume on the contrary that there was an odd length cycle. We can assume without loss of generality that the cycle starts with a vertex in X , and call it v_1 . In each step, it goes to a vertex in Y , and then X , then Y and so on. After $2n + 1$ steps, it will end with a vertex in X v_{2n+1} , at which point it forms a cycle by connecting with v_1 . However, this cannot happen since for a graph to be bipartite, there can be no edges that connect two vertices in X . This is a contradiction, so there are no odd length cycles.

Back: Assume first that G is connected (otherwise, apply the following argument to each connected component separately). Let r be any vertex. Let X be the set of vertices at an even distance from r , and let Y be the set of vertices at an odd distance from r .

We claim that every edge in the graph joins a vertex in X to a vertex in Y . Assume on the contrary that there exists an edge (u, v) such that either $u, v \in X$ or $u, v \in Y$. Let P_u and P_v be shortest paths from r to u and v , respectively. Let w be the last vertex common to both paths. Then the portions of P_u and P_v from w to u and from w to v are internally disjoint. Together with the edge (u, v) , they form a cycle.

We now consider two cases:

- **Case:** $u, v \in X$. Since both u and v are an even distance from r , the paths from w to u and from w to v have the same parity. Let their lengths be $2a$ and $2b$, or $2a + 1$ and $2b + 1$. In either case, their sum is even. Adding the edge (u, v) gives a cycle of odd length. This contradicts the assumption that G has no odd cycles.
- **Case:** $u, v \in Y$. Since both u and v are an odd distance from r , the paths from w to u and from w to v again have the same parity, so their sum is even. Adding the edge (u, v) gives a cycle of odd length, contradicting the assumption that G has no odd cycles.

Since both cases lead to contradictions, every edge joins a vertex in X to a vertex in Y . Therefore, (X, Y) is a bipartition of G , so G is bipartite. $\square$

16. **Claim:** Every tree with $n \geq 2$ vertices has at least two vertices of degree 1.

Proof. We will use the result that an n -vertex tree has to have $n - 1$ edges from Question 18 to answer this question. We use proof by contradiction and the Handshaking Lemma to prove this statement. Assume on the contrary that it's not the case that at least two vertices have degree 1. We have two cases now:

- **Case: No vertices of degree 1.** We have $\deg(v) \geq 2$ for all vertices. Thus, we have $\sum_{v \in G} \deg(v) \geq 2(n) = 2n$. By the Handshaking Lemma, we also know this sum is $2|E| = 2(n - 1)$, so we get $2(n - 1) \geq 2n$, or $-2 \geq 0$, which is clearly impossible.
- **Case: One vertex of degree 1.** In this case, we have $\sum_{v \in G} \deg(v) \geq 2n - 1$, and thus our final inequality is $2n - 1 \geq 2(n - 1)$, or $-1 \geq 2$, which again is clearly impossible.

Since both cases are impossible, it follows that our assumption is wrong. Therefore, the claim is true. $\square$

17. **Claim:** The two stated properties of a tree (connected with no cycles / unique path between any two vertices) are equivalent.

Proof. Front: Since the graph is connected with no cycles, we can reach any vertex from any vertex without repeating any vertices. This means that there's a path between any two vertices. We also know that this path is unique because if it wasn't, then we could walk from u to v along one path and from v to u along another, forming a cycle, which would be a contradiction.

Back: Since there's a unique path between any two vertices, the graph is connected. Assume on the contrary that the graph has cycles. Then, by choosing any two distinct vertices on the cycle, there are two different paths going between them, each going different ways on the cycle. This contradicts our assumption that there's a unique path, so there are no cycles. $\square$

18. **Claim:** A tree with n vertices has $n - 1$ edges.

Proof. Let $E(n)$ be the number of edges on a tree with n vertices. We'll use induction to prove that $E(n) = n - 1$:

- **Base case:** $n = 1$. This graph clearly has 0 edges, since there's no other vertex to connect to. We have $E(1) = 0 = 1 - 1$ as desired.
- **Inductive Hypothesis:** Assume $E(n) = n - 1$.

- **Inductive Step:** $E(n+1) = n$. First, find a leaf of the graph. To prove a leaf exists, consider the longest path in the tree, $v_0, v_1, \dots, v_m$. We can't connect v_0 to any other vertex without making a longer path (violating our assumption that this is the longest path), or connecting to vertices in the path (which would create a cycle, which isn't allowed). Thus, v_0 only has one neighbor, v_1 , and so is a leaf.

Now, remove v_0 and the edge (v_0, v_1) . We still have a tree because the remaining graph is still connected and doesn't have cycles. This tree specifically has n vertices. From our inductive hypothesis, we know that our new truncated tree has $n-1$ edges. Now adding v_0 and (v_0, v_1) back in, we've added one vertex and one edge, so $E(n+1) = n$ as desired.

□

19. **Claim:** If G is a connected graph with n vertices, then G has at least $n-1$ edges.

Proof. We'll use induction once again. Let $E(G, n)$ be the number of edges a connected graph G with n vertices has. We want to prove that $E(G, n) \geq n-1$ for all n, G :

- **Base case:** $n = 1$. This graph is fully connected no matter how many edges it has, including 0 edges, so we have $E(n) \geq 0 = 1-1$ as desired.
- **Inductive hypothesis:** Assume that $E(G, n) \geq n-1$ for all G, n .
- **Inductive step:** $E(G, n+1) \geq n$. If we remove an edge in graph G , it might split the graph into two connected components. Let's call these components X and Y . X has a vertices and Y has b vertices, so $n+1 = a+b$. Note that $E(X, a) \geq a-1$ and $E(Y, b) \geq b-1$, so we have:

$$\begin{aligned} E(G, n+1) &= E(X, a) + E(Y, b) + 1 \\ &\geq (a-1) + (b-1) + 1 \\ &= (a+b) - 1 \\ &= (n+1) - 1 \\ &= n, \end{aligned}$$

as desired.

□

20. **Trees with n vertices and diameter d .** To construct such a tree, create a path graph P_{d+1} with vertices $v_0, v_1, \dots, v_d$. Then, attach the other $n-d-1$ vertices as leaves to v_0 . This creates a tree with diameter d .

21. **Functions on a directed graph adjacency list.** See file `p21.py` for the code for this problem. It also includes test cases on a sample adjacency list created at the beginning of the script.
22. **Random Walk code** See file `p22.py` for the code for this problem. It simulates a random walk and creates a table consisting of how often a vertex was visited during the random walk. It is in percentages, so the sum of the entries in the table add up to 100.
23. **Simulation on Lollipop Graph:** Using the code in `p22.py`, I simulated a random walk on the Lollipop Graph with 10 vertices and got the following results:

1	13%
2	13%
3	13%
4	13%
5	17%
6	7%
7	7%
8	7%
9	7%
10	3%

Table 2: Frequency table of vertices visited